

Back

sec/1.introduction.tex

Source

Split

Rendered

Sync

Compile

Compile All

Files

Upload

.omx

archive

docs

architecture-diagrams.md

markdown-draft.md

notes.md

review-report-v2.md

review-report-v3.md

review-report-v4.md

review-report-v5.md

review-report-v6.md

review-report.md

wiley-spe-conversion-guid...

figures

sec

1.introduction.tex

2.related-work.tex

3.design.tex

4.implementation.tex

5.evaluation.tex

6.discussion.tex

7.limitations.tex

8.conclusion.tex

wiley

images

lettersp.sty

main.pdf

main.tex

NJDnatbib.sty

prepare-submission-archive...

README.md

references.bib

USG.cls

verify.sh

wileyNJD-Chicago-lasttwo.bst

wileyNJD-Chicago.bst

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

\section{Introduction}

Academic writing is a complex document engineering process. In disciplines that use \LaTeX{}, a manuscript is not merely a linear text document: it is a project composed of source files, bibliographic databases, figures, style templates, experimental code, compiler logs, reviewer feedback, and final PDF artifacts. Researchers must coordinate editing, citation management, compilation, revision tracking, and collaboration across these interconnected components.

The emergence of large language models (LLMs)~\cite{openai2023gpt4, anthropic2024claude, openai2024gpt4o} has introduced new possibilities for AI-assisted writing. Tools such as ChatGPT, Claude, Gemini, and specialized academic assistants can generate, paraphrase, summarize, and critique scholarly text. However, general-purpose chat interfaces treat writing as an isolated text-generation task. They separate model output from the project context in which scholarly artifacts must be edited, cited, compiled, and reviewed. This separation creates three practical problems:

\begin{enumerate}\item \textbf{Context fragmentation.} Researchers must repeatedly copy manuscript sections into the assistant, increasing friction and the risk of stale or incomplete prompts. When the manuscript changes, chat context diverges from source truth.\item \textbf{Unaudited edits.} When assistants can directly modify files, or when changes are manually copied into a source tree, it becomes difficult to track what was changed, by whom, and why. This undermines the audit trail essential for scholarly integrity.\item \textbf{Unverifiable artifacts.} Generated text may introduce

PDF

DIFF

A- 14px A+

## 1 Introduction

Academic writing is a complex document engineering process. In disciplines that use \LaTeX{}, a manuscript is not merely a linear text document: it is a project composed of source files, bibliographic databases, figures, style templates, experimental code, compiler logs, reviewer feedback, and final PDF artifacts. Researchers must coordinate editing, citation management, compilation, revision tracking, and collaboration across these interconnected components.

The emergence of large language models (LLMs) [openai2023gpt4, anthropic2024claude, openai2024gpt4o] has introduced new possibilities for AI-assisted writing. Tools such as ChatGPT, Claude, Gemini, and specialized academic assistants can generate, paraphrase, summarize, and critique scholarly text. However, general-purpose chat interfaces treat writing as an isolated text-generation task. They separate model output from the project context in which scholarly artifacts must be edited, cited, compiled, and reviewed. This separation creates three practical problems:

- Context fragmentation.** Researchers must repeatedly copy manuscript sections into the assistant, increasing friction and the risk of stale or incomplete prompts. When the manuscript changes, chat context diverges from source truth.
- Unaudited edits.** When assistants can directly modify files, or when changes are manually copied into a source tree, it becomes difficult to track what was changed, by whom, and why. This undermines the audit trail essential for scholarly integrity.
- Unverifiable artifacts.** Generated text may introduce hallucinated references, missing citation keys, or \LaTeX{} changes that fail to compile. Liang et al. [liang2024llmwriting] documented widespread use of LLMs in scientific manuscripts and highlighted significant risks of hallucinated references, underscoring the verification gap. These are document engineering failures that text-generation quality metrics do not capture.

AI Assistant

Chat Skills RAG Review Citations Anti-AI Pipeline

### Pipeline 2.0

Composable multi-stage workflows with typed executors

Paper Pipeline — Polish → Review → Revise → Compile

polish ai

review ai

review-checkpoint human

revise ai

citation-check citation

compile compile

Start Pipeline